

Durable, Energy-Efficient Lighting for Public Housing

A prototype light fixture gets put through its paces at the Chicago Housing Authority and comes out shining.

by Robert L. Wendt
and Clark Lundell

Most Americans take well-lit hallways and stairwells for granted. Residents at the Chicago Housing Authority (CHA) in Chicago, Illinois, haven't done so for years. Last October, though, that situation began to change when six uniquely designed prototype lighting fixtures were installed for field testing.

These fixtures were the result of efforts by industrial design students at Auburn, Alabama-based Auburn University and researchers at Oak Ridge National Laboratory. "We've been able to go beyond what's offered commercially and come up with a design that specifically addresses issues like maintenance, safety, and efficiency," says Dave Anderson, head of technical services for CHA. While they were designed to meet the specific needs of CHA, the light fixtures are expected to have broad applicability to other housing authorities, schools, and public buildings where rough use is projected.

The Problem

Common-area lighting at CHA poses a critical problem, as maintenance staff struggle to maintain acceptable light levels against huge odds. Many structures retain the original incandescent fixtures that were installed in the 1950s and '60s. Years of use and exposure to the elements have deteriorated many fixtures and left others virtually inoperative. Accessible light bulbs in these fixtures are subject to theft. When the fixtures are working, they are often subjected to vandalism that is intended to put them out of service. A number of replacement techniques have been tried, but none has been fully successful.

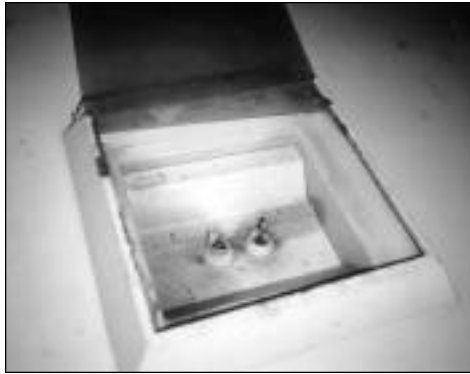
CHA needs durable, energy-efficient lighting that would:

- provide light at levels required by applicable codes and standards;
- withstand the adverse conditions imposed by the environment in which it is located, including summer

temperatures of 105°F, winter temperatures of -30°F, wet conditions, and exposure to ice, snow, and various deicers;

- provide adequate light and retain a reasonable appearance while withstanding exposure to spray paint, gunshots, cigarette lighters, all forms of mechanical abuse, attempted theft, and attempts to disconnect the fixtures from the power source;
- use high-efficiency artificial light sources or daylight;
- be acceptable to the residents of the housing project;
- provide a reasonable first cost that would encourage installation by housing authorities and other potential users;
- provide the lowest life-cycle costs, including first cost, maintenance costs, energy costs, and life expectancy; and
- be easily maintained by people with a wide range of skills, from trained residents to electricians.

Auburn's goals for this project were to install and test a limited number of prototype fixtures under real-world conditions. Based on prototype testing, refinements will be made to the design, and working drawings will be produced to enable CHA and others to acquire refined fixtures for installation in other housing projects.



The existing incandescent fixtures hold two 60W bulbs. The lens that covers the bulbs is easily removed or broken.

The Prototype

After evaluating many original and thoughtful solutions to the lighting problem, CHA chose to develop and test a prototype fixture made of 16-gauge steel with a white powder coat finish. The fixture is 13 inches square, with slanted sides to help reflect light. Two 18W compact fluorescent bulbs with electronic ballasts provide 20% more light than do the two 60W incandescent bulbs in the current fixtures. The dome-shaped lens is fabricated of milky polycarbonate. The lens, reflective enclosure, and brighter bulbs of the prototype fixtures provide significantly more light than do the deteriorated, existing fixtures.

The new fixtures fit within an existing standard 13 inch x 13 inch metal box recessed into the underside of the concrete slab that forms the floors/ceilings of the housing projects, minimizing

its exposure. Surface mounting techniques are also being investigated to broaden the potential application of the fixture.

Many variables were considered in all phases of design development, including light level, energy consumption, maintenance, safety, function, and cost. For example,

the fixture was to be as maintenance-friendly as possible. The prototype keeps all the parts of the fixture together when a light bulb needs to be changed. Screws holding the lens to the mounting remain in place, and the lens drops down several inches and out of the way on a cable so a maintenance worker can change a light bulb without having to hold, or keep track of, the various parts of the fixture. In addition, the compact fluorescent bulbs will last about ten times longer than the current fixture's incandescent bulbs.

Prototype Testing

During the development of the prototype, various elements of the design were subjected to testing to learn how they would perform under field conditions. The polycarbonate lens, which is the primary exposed element in the design, was tested to determine how well it would withstand various forms of vandalism, including gunshots, battering, spray paint, and burning with cigarette lighters.

Gunshots from typical handguns did not penetrate the lens, but did not destroy it. Close-range

gunshots could theoretically be used to break the light bulbs and render the fixture inoperative. However, observation of vandalized light fixtures at CHA suggested that gunshots are not a major threat for these fixtures. Street lights and other remote light sources were more likely targets for gunshots.

Battering light fixtures with a blunt object was somewhat more popular. An aluminum baseball bat was used to test the resistance of the lens to this form of vandalism. In this contest the polycarbonate lens outperformed the bat, surviving the encounter with only some minor scuffing, while the bat was visibly dented.

Vandalism with spray paint is very common in public housing. The polycarbonate lens and metal frame can be messed up aesthetically with this form of vandalism. Films and coatings are available that could provide a non-stick



The existing light fixtures commonly serve as canvases for graffiti.



The prototype fixture features a domed polycarbonate lens and tamper-resistant fasteners.

surface, but most of them would add significantly to the initial cost of the fixture. As a low-cost alternative, a spray-on wax furniture polish was applied to three of the prototypes in the field test to see how it would perform. This wax should break the adhesion of the paint to the lens, causing the paint to form droplets that will fall off the fixture due to their weight. Exposure to heat, moisture, and wind may reduce its effectiveness over time.

Based on field observations, the most damaging form of vandalism against light fixtures is to burn a hole in the lens with a cigarette lighter and then break the bulb with a stick. The polycarbonate lens of the prototype is vulnerable to this form of vandalism. However, unlike that of other com-



Existing vandal-resistant fixtures were put out of commission by burning a hole in the lens and then breaking the bulb with a stick.

mercially available fixtures, the configuration of the prototype offers the opportunity to add a second level of protection, such as a fine mesh metal screen or laminated safety glass. These materials would provide a translucent, physical barrier to protect the bulbs, and they are less sensitive to fire. This added protection would increase the cost of the fixture, but it is not expected to be required in all applications. The performance of the prototypes used in the field test will help to determine if this additional protection should be provided as a standard or optional feature. Through the first two months of the field test, no vandalism of any kind occurred, and no conclusions could be reached about the prototype's vandalism survival rate.

Another area of concern was the performance of the compact fluorescent bulbs under the extreme cold associated with Chicago winters. One bulb manufacturer's data claimed performance to -20°F . To confirm this information, candidate bulbs were placed in a residential freezer and operated for several days at about 0°F . No degradation of performance was observed. It is believed that the actual temperature surrounding the bulbs will not drop below 0°F , since they will operate continuously in actual service. Field testing during January and February 2000—typically the coldest months in Chicago—confirmed satisfactory performance under actual win-

ter conditions. However, the temperature extremes during this period were less than in a typical year, with air temperatures remaining at or above 0°F throughout the period.

Monitoring and Payback

Last October, six prototypes were installed in one of the 17-story Robert Taylor Homes.

Weekly monitoring of the fixtures was conducted for the first several months, followed by monthly checks. Through the end of 1999, the fixtures remained as installed. No vandalism or operational problems had

occurred. During this period, temperatures remained mild, and ongoing renovation of the building had caused the power to the fixtures to be shut off several times, so prototype performance during low-temperature conditions and length of bulb life are not yet known.

The existing fixtures in CHA's high-rise housing contain two 60W incandescent bulbs and operate continuously. Each fixture consumes about 1,050 kWh, or about \$76 of power, per year at CHA's \$.072/kWh rate. The new fixtures contain two 18W compact fluorescent bulbs, which, along with electronic ballasts, will consume about 40W per fixture. These new fixtures will consume about 350 kWh, or about \$25 of power, per year. With an energy cost savings of \$51 per year, the new fixtures are expected to have a simple payback of about 2.5 years based on energy savings alone. Fixture cost is estimated to be \$100, and installation cost is estimated

to be \$25–\$35, per unit. In-kind replacements for the original fixtures cost \$30–\$40 less than the prototypes, but since these are intended for normal residential use, not rough use, comparing the cost of the fixtures may not be valid.

Another of the fixtures' impressive features is the dramatically reduced maintenance costs for changing bulbs. The existing incandescent bulbs can last up to 1,000 hours in service. However, they seldom do, because they are often stolen. The new fixtures, with their more durable enclosure and their compact, push-in base, fluorescent bulb, which is less desirable to thieves, are expected to achieve their full projected life of 10,000 hours in service. This will cut bulb-changing maintenance by a factor of at least ten, which in turn will reduce maintenance costs by about \$75 per year per fixture. Factoring in this figure, the new fixtures are expected to have a simple payback of about 1.8 years based on maintenance savings alone. Combining both energy and maintenance savings yields a simple payback of just 1.1 years.

Another factor that has been hard to quantify is long-term durability. Because the fixtures have been specifically designed to withstand the hostile environment of severe weather extremes and periodic exposure to vandalism, replacement costs are projected to be lower than other alterna-



A Chicago Housing Authority staff electrician is lighting up a building's entryway with one of the prototype lighting fixtures.



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Researchers tested the prototype fixture's vulnerability to abuse with a baseball bat. In the bat-versus-fixture contest, the polycarbonate lens outperformed the bat, surviving the encounter with only some minor scuffing, while the bat was visibly dented.

means to determine whether this will prove to be the case.

CHA officials and residents have been so impressed with the quality of the prototype fixtures that they have installed 200 more of them. In addition, CHA is making plans to purchase 1,600 additional fixtures for ten more high-rise multifamily buildings later this year. CHA is not the only organization to be pleased with the design. Last October, Auburn University won a special design award from the Association of Energy Engineers, Chicago chapter, for the research, design, and demonstration of a durable, energy-efficient lighting fixture.

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This project was funded by the Department of Energy's Office of Buildings Technology, State and Community Programs and managed by the Oak Ridge National Laboratory. The Chicago Housing Authority was the client. CHA provided the problem to be solved, evaluated the design, and provided the site for the field test. The department of Industrial Design at Auburn University provided the principal investigator and coordinated the student design effort. Argonne National Laboratory served as liaison and facilitator. Industry advisors on the project included Lithonia Lighting, Southco Fasteners, and Prolight. They provided technical review and input and assistance in fabricating the prototypes.

tive fixtures that are currently available to CHA. Long-term exposure to this hostile environment is the only

pter, for the research, design, and demonstration of a durable, energy-efficient lighting fixture.